Value-at-Risk Model Risk

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Abstract

Large banks assess their regulatory capital for market risk using complex, firm-wide Value-at-Risk (VaR) models. In their 'bottom-up' approach to VaR there are many sources of model risk. A recent amendment to banking regulations requires additional market risk capital to cover all these model risks but, as yet, there is no accepted framework for computing such an add-on. We introduce a top-down approach to quantifying VaR model risk in a rigorous statistical framework and derive a corresponding adjustment to regulatory capital that is relatively straightforward to implement.

Key Words: Basel II; Maximum entropy; Model risk; Quantile; Risk capital; Value-at-Risk (VaR)

JEL Codes: C1, C19, C51, G17, G28

1 Introduction

The term 'model risk' is commonly applied to encompass various sources of uncertainty in statistical models. Following Cairns (2000) we distinguish two sources of model risk:[1]

- *Model choice*, i.e. inappropriate assumptions about the form of the statistical model for the random variable;
- *Parameter uncertainty*, i.e. estimation error in the parameters of the chosen model.[2]

This paper focuses on the model risk of quantile risk assessments with particular reference to 'Value-at-Risk' (VaR) estimates, which are derived from quantiles of portfolio profit and loss distributions. VaR corresponds to an amount that could be lost, with a specified probability, if the portfolio remains unmanaged over a specified time horizon. It has become the global standard for assessing risk in all types of financial firms: in fund management, where portfolios with long-term VaR objectives are actively marketed; in the treasury divisions of large corporations, where VaR is used to assess position risk; and in insurance companies, who measure underwriting and asset management risks in a VaR framework. But most of all, banking regulators remain so confident in VaR that its application to computing market risk capital for banks, used since the 1996 amendment to the Basel I Accord,[3] will soon be extended to include stressed VaR under an amended Basel II and the new Basel III Accords.[4]

The finance industry's reliance on VaR has been supported by decades of academic research. Especially during the last ten years there has been an explosion of articles published on this subject – see Christoffersen (2009) for a survey. A popular topic is the introduction of new VaR models,[5] and another very prolific strand of literature focuses on testing their accuracy.[6] However, the stark failure of many banks to set aside sufficient capital reserves during the banking crisis of 2008 sparked an intense debate on using

[1]There is no consensus on the sources of model risk. For instance, Cont (2006) points out that both these sources could be encompassed within a universal model, and Kerkhof et al. (2010) distinguish 'identification risk' as an additional source.

[2]This includes sampling error (parameter values can never be estimated exactly because only a finite set of observations on a random variable are available) and optimization error (e.g. different numerical algorithms typically produce slightly different estimates based on the same model and the same data).

[3]See Basel Committee on Banking Supervision (1996).

[4]See Basel Committee on Banking Supervision (2009).

[5]Historical simulation (Hendricks, 1996) is the most popular approach amongst banks (Perignon and Smith, 2010) but data-intensive and prone to pitfalls (Pritsker, 2006). Other popular VaR models assume normal risk factor returns with the RiskMetrics covariance matrix estimates (RiskMetrics, 1997). More complex VaR models are proposed by Hull and White (1998b), Mittnik and Paolella (2000), Christoffersen et al. (2001), Ventner and de Jongh (2002), Angelidis et al. (2004), Hartz et al. (2006), Kuan et al. (2009) and many others.

[6]The coverage tests introduced by Kupiec (1995) are favoured by banking regulators, and these are refined by Christoffersen (1998), Christoffersen et al. (2001) and Christoffersen and Pelletier (2004). However Berkowitz et al. (2010) demonstrate that more sophisticated tests such as the conditional autoregressive test of Engle and Manganelli (2004) may perform better.

VaR models for the purpose of computing the market risk capital requirements of banks. Turner (2009) is critical of the manner in which VaR models have been applied and Taleb (2007) even questions the very idea of using statistical models for risk assessment. Despite the warnings of Turner, Taleb and other critics of VaR models,[7] most financial institutions continue to employ them as their primary tool for market risk assessment and economic capital allocation.

For internal, economic capital allocation purposes VaR models are commonly built using a 'bottom-up' approach. That is, VaR is first assessed at an elemental level, e.g. for each individual trader's positions, then is it progressively aggregated into desk-level VaR, and VaR for larger and larger portfolios, until a final VaR figure for a portfolio that encompasses all the positions in the firm is derived. This way the traders' limits, and risk budgets for desks and broader classes of activities can be allocated within a unified framework. However, this bottom-up approach introduces considerable complexity to the VaR model for a large bank. Indeed, it could take more than a day to compute the full (often numerical) valuation models for each product over all the simulations in a VaR model. Yet, for regulatory purposes VaR must be computed at least daily, and for internal management intra-day VaR computations are frequently required.

To reduce complexity in the internal VaR system simplifying assumptions are commonly used, in the data generation processes for financial assets and interest rates and in the valuation models used to mark complex products to market every day. For instance, it is very common to apply normality assumptions in VaR models, along with lognormal, constant volatility approximations for exotic options prices and sensitivities.[8] Of course, there is conclusive evidence that financial asset returns are not well represented by normal distributions. However, the risk analyst in a large bank may be forced to employ this assumption for pragmatic reasons.

Another common choice is to base VaR calculations on simple historical simulation. Many large commercial banks have legacy systems that are only able to compute VaR using this approach, commonly basing calculations on at least 3 years of daily data for all traders' positions. Thus, some years after the credit and banking crisis vastly overinflated VaR estimates could still be produced by these models, even though markets have returned to normal. The implicit and simplistic assumption that history will repeat itself with certainty – that the banking crisis will recur within the risk horizon of the VaR model – may well seem absurd to the analyst, yet he is constrained by the legacy system to compute VaR using simple historical simulation.

[7]Even the use of a quantile has been subject to criticism, as it is not necessarily sub-additive, and related metrics such as conditional VaR may be preferred. See Beder (1995) and Artzner et al. (1999).

[8]Indeed, model risk frequently spills over from one business line to another, e.g. normal VaR models are often employed in large banks simply because they are consistent with the geometric Brownian motion assumption that is commonly applied for option pricing and hedging.

Thus, financial risk analysts are often constrained to employ a model that does not comply with their beliefs about the data generation processes for financial returns, and the data that they think is inappropriate.[9] As a result they have long recognized model risk due to model choice and parameter uncertainty. In the literature, Derman (1996), Simons (1997), Crouhy et al. (1998), Green and Figlewski (1999), Kato and Yoshiba (2000) and Rebonato (2001) all identify these two main causes of model risk in finance. However, in those papers a formal, quantitative definition of model risk that allows its assessment is elusive.

Surprizingly few papers deal explicitly with VaR model risk. Early work by Jorion (1996) and Talay and Zheng (2002) investigated sampling error and Brooks and Persand (2002) assessed the effect of applying different GARCH models to estimate VaR. At the time of writing the only other published paper in this important area is by Kerkhof et al. (2010). It addresses the VaR model risk stemming from both model choice and parameter uncertainty and, by quantifying the adjustment to VaR that is necessary for normal i.i.d. and GARCH models to pass regulatory backtests, the authors derive an incremental market risk capital charge (henceforth simply called the risk capital 'add-on') to cover VaR model risk. This issue is very important because recent revisions to the Basel II market risk framework include the requirement that banks set aside additional reserves to cover all sources of model risk in the internal models used to compute the market risk capital charge.[10]

This paper introduces a new framework for measuring quantile model risk with an elegant, intuitive and practical method for computing the risk capital add-on to cover VaR model risk. In the model development section we employ the general 'quantile' rather than 'VaR' terminology because our methodology has potential applications to many areas, not only to finance.[11] In addition to the computation of a model risk 'add-on' for a given VaR model and given portfolio,[12] our approach can be used to assess which, of the available VaR models, has the least model risk (relative to a given portfolio). Similarly, given a specific VaR model, our approach can assess which portfolio has the least model risk.

[9]Banking regulators recommend 3-5 years of data for historical simulation and require at least 1 year of data for constructing the covariance matrices used in other VaR models.

[10]See Basel Committee on Banking Supervision (2009), Section IV.

[11]Quantile estimation is commonly applied to a variety of disciplines: a survey of their applictions to insurance, actuarial science, hydrology and several other fields as well as finance is given by Reiss and Thomas (1997). The probability of the occurrence of extreme events is of prime interest for actuaries, where heavy-tailed distributions are used to model large claims and losses (for instance, see Matthys et al., 2004); heavy-tailed distributions are frequently used for quantile estimation in hydrology and climate change (Mkhandi et al. (1996), Katz et al. (2002)) in statistical process control for computing capability indices (Anghel, 2001), for measuring efficiency (Wheelock and Wilson, 2008) and for reliability analysis (Unnikrishnan Nair and Sankaran, 2009). The uncertainty surrounding quantile-based risk assessments in these areas has long been recognised (see Peng and Qi, 2006).

[12]A portfolio could be at any level of granularity, e.g. an individual trader's positions, or all positions in a desk, or the entire daily P&L of a bank.

In the following: the benchmark for assessing model risk is discussed in Section 2; Section 3 gives a formal definition of quantile model risk and outlines a framework for its quantification. We present a statistical model for the probability $\hat{\alpha}$ that is assigned, under the benchmark distribution, to the α quantile of the model distribution. Our idea is to endogenize model risk by using a distribution for $\hat{\alpha}$ to generate a distribution for the quantile. The mean of this model-risk-adjusted quantile distribution detects any systematic bias in the model's α quantile, relative to the α quantile of the benchmark distribution. A suitable quantile of the model-risk-adjusted distribution determines an uncertainty buffer which, when added to the bias-adjusted quantile gives a model-risk-adjusted quantile that is no less than the α quantile of the benchmark distribution at a pre-determined confidence level, this confidence level corresponding to a penalty imposed for model risk; Section 4 presents an empirical example on the application of our framework to VaR model risk, in which the degree of model risk is controlled by simulation; Section 5 summarizes and concludes.

2 The Benchmark

Model risk in finance has been approached in two different ways: examining all feasible models and evaluating the discrepancy in their results, or specifying a benchmark model against which model risk is assessed. Papers on the quantification of valuation model risk in the risk-neutral measure exemplify each approach: Cont (2006) quantifies the model risk of a complex product by the range of prices obtained under all possible valuation models that are calibrated to market prices of liquid (e.g. vanilla) options; Hull and Suo (2002) define model risk relative to the implied price distribution, i.e. a benchmark distribution implied by market prices of vanilla options. In the context of VaR model risk the benchmark approach, which we choose to follow, is more practical than the former.

Some authors identify model risk with the departure of a model from a 'true' dynamic process: see Branger and Schlag (2004) for instance. Yet, outside of an experimental or simulation environment, we never know the 'true' model for sure. In practice, all we can observe are realisations of the data generation processes for the random variables in our model. It is futile to propose the existence of a unique and measurable 'true' process because such an exercise is beyond our realm of knowledge.

However, we can observe a maximum entropy distribution (MED). This is based on a 'state of knowledge', i.e. no more and no less than the information available regarding the random variable's behaviour. This information includes the observable data that are thought to be relevant plus any subjective beliefs. Shannon (1948) defined the entropy of a probability density function $g(x)$, $x \in \mathcal{R}$ as

$$H(g) = -\mathrm{E}_g[\log g(x)] = -\int_{\mathcal{R}} g(x) \log g(x) dx.$$

This is a measure of the uncertainty in a probability distribution and its negative is a measure of information.[13] The maximum entropy density is the function $f(x)$ that maximizes $H(g)$, subject to a set of conditions on $g(x)$ which capture the testable information that is available.[14] The criterion here is to be as vague as possible (i.e. to maximize uncertainty) given the constraints imposed by the testable information. This way, the maximum entropy distribution (MED) represents no more (and no less) than this information. If the testable information consists only of a historical sample on X of size n then, in addition to the normalization condition, there are n conditions on $g(x)$, one for each data point. In this case, the MED is just the empirical distribution based on that sample. Otherwise, the testable information consists of fewer conditions, which capture only that sample information which is thought to be relevant, and any other conditions imposed by subjective beliefs about the MED.

From the banking regulator's perspective what matters is not the ability to aggregate and disaggregate VaR in a bottom-up framework, but the adequacy of a bank's total market risk capital reserves, which are derived from the aggregate market VaR. Therefore, regulators only need to define a benchmark VaR model to apply to the bank's aggregate daily profit and loss (P&L). This model will be the MED of the regulator, i.e. the model that best represents the regulator's state of knowledge regarding the accuracy of VaR models. Our recommendation is that banks assess their VaR model risk by comparing their aggregate VaR figure, which is typically computed using the 'bottom-up' approach, with the VaR obtained using the regulator's MED in a 'top-down' approach, i.e. calibrated directly to the bank's aggregate daily trading P&L. Typically this P&L contains marked-to-model prices for illiquid products, in which case their valuation model risk is not quantified in our framework.

Following the theoretical work of Shannon (1948), Zellner (1977), Jaynes (1983) and many others it is common to assume the testable information is given by a set of moment functions derived from a sample, in addition to the normalization condition. When only the first two moments are deemed relevant, the MED is a normal distribution (Shannon,

[13] For instance, if g is normal with variance σ^2, $H(g) = \frac{1}{2}(1+\log(2\pi)+\log(\sigma))$, so the entropy increases as σ increases and there is more uncertainty and less information in the distribution. As $\sigma \to 0$ and the density collapses the Dirac function at 0, there is no uncertainty but $-H(g) \to \infty$ and there is maximum information. However, there is no universal relationship between variance and entropy and where their orderings differ entropy is the superior measure of information. See Ebrahimi, Maasoumi and Soofi (1999) for further insight.

[14] A piece of information is *testable* if it can be determined whether F is consistent with it. One of piece of information is always a normalization condition.

1948). More generally, when the testable information contains the first N sample moments, $f(x)$ takes an exponential form. This is found by maximizing entropy subject to the conditions $\mu_n = \int_{\mathcal{R}} x^n g(x)dx$ for $n = 0, \ldots, N$, where $\mu_0 = 1$ and μ_n $(n = 1, ..., N)$ are the moments of the distribution. The solution is $f(x) = \exp\left(-\sum_{n=0}^{n=N} \lambda_n x^n\right)$ where the parameters $\lambda_0, \ldots \lambda_n$ are obtained by solving the system of non-linear equations $\mu_n = \int x^n \exp\left(-\sum_{n=0}^{n=N} \lambda_n x^n\right) dx$, for $n = 0, \ldots, N$.

Rockinger and Jondeau (2002), Wu (2003) and several others have applied a four-moment MED to various econometric and risk management problems. Park and Bera (2009) and Chan (2009a) apply a four-moment MED to GARCH models and Chan (2009b) extends this to the computation of VaR. But, perhaps surprizingly, none of these papers consider the tail weight that is implicit in the use of a four-moment MED. In fact, moment-based MEDs are only well-defined when N is even. For any odd value of N there will be an increasing probability weight in one of the tails. Also, the four-moment MED has lighter tails than a normal distribution, due to the presence of the term $\exp[-\lambda_4 x^4]$ with non-zero λ_4 in $f(x)$. Indeed, the more moments included in the conditions, the thinner the tail of the MED. But financial asset returns are typically heavy-tailed and it is likely that this property will carry over to a bank's aggregate daily P&L. Hence, we cannot advocate the use of moment-based MEDs.

Nevertheless, there are advantages in choosing a parametric MED for the regulator's benchamrk. VaR estimates are quantiles of a forward-looking P&L distribution, but to base model parameter estimates entirely on historical data limits beliefs about the future to what has been experienced in the past. Parametric distributions are frequently advocated for VaR estimation because the parameter values estimated from historical data may be changed subjectively to accomodate beliefs about the future PL distribution.

The flexible class of generalized beta generated (GBG) distributions introduced by Alexander et al. (2010) have three parameters that offer direct control over peakness, skew and relative tail weights. The GBG distribution is defined by replacing the uniform $U[0, 1]$ distribution in the probability integral transform by a generalized beta distribution (of the first kind, introduced by McDonald, 1984) denoted $\mathcal{GB}(a, b, c)$. This may be characterized by its density function

$$f_{\mathcal{GB}}(u; a, b, c) = B(a, b)^{-1}[cu^{ac-1}(1 - u^c)^{b-1}], \quad 0 < u < 1, \ \ a, b, c > 0. \tag{1}$$

Given any continuous parent distribution $F(x)$, $x \in \mathcal{R}$ with density $f(x)$, X has a GBG distribution when the probability transformed variable $U = F(X)$ has density (1). Then the distribution of X may be characterised by its density function:

$$f_{\mathcal{GBG}}(x; a, b, c) = B(a, b)^{-1} f(x)[cF(x)^{ac-1}(1 - F(x)^c)^{b-1}], \ x \in \mathcal{R}. \tag{2}$$

GBGs are MEDs under just three, fairly general shape constraints, so an advantage of using a GBG for the benchmark VaR model is that the parameter values that are estimated from historical data may be adjusted subjectively, to reflect specific beliefs about the peakness, skew and relative tail weights of future P&L. Another reason for our interest in GBG distributions is that they will become central to our analysis in Section 3.

Following the study by Berkowitz and O'Brien (2002) on the aggregate performance of VaR models for large commercial banks, regulators may prefer to specify a benchmark model for VaR, calibrated to aggregate daily P&L, from the familiar GARCH class. Berkowitz and O'Brien found that most 'bottom-up' internal VaR models produced VaR estimates that were too large, and insufficiently risk-sensitive, compared with the GARCH VaR estimates that were derived using only the aggregate daily P&L. Filtered historical simulation (FHS) of aggregate daily P&L would be another popular alternative, especially when combined with a volatility filtering that increases its risk sensitivity: see Barone-Adesi et al. (1998) and Hull and White (1998a). Alexander and Sheedy (2008) demonstrated empirically that GARCH volatility filtering combined with histroical simulation can produce very accurate VaR and conditional VaR estimates, even at extreme quantiles. By contrast, the standard historical simulation approach failed many of their backtests.

3 Quantile Model Risk

The α quantile of a continuous distribution F of a real-valued random variable X with range $\mathcal{R}$ is denoted[15]

$$q_\alpha^F = F^{-1}(\alpha). \tag{3}$$

In our statistical framework F is identified with the unique MED based on a state of knowledge $\mathcal{K}$ which contains all testable information on F. We characterise a statistical model as a pair $\{\hat{F}, \hat{\mathcal{K}}\}$ where $\hat{F}$ is a distribution and $\hat{\mathcal{K}}$ is a filtration which encompasses both the model choice and its parameter values. The model provides an estimate $\hat{F}$ of F, and uses this to compute the α quantile. That is, instead of (3) we use

$$q_\alpha^{\hat{F}} = \hat{F}^{-1}(\alpha). \tag{4}$$

Quantile model risk arises because $\{\hat{F}, \hat{\mathcal{K}}\} \neq \{F, \mathcal{K}\}$. Firstly, $\hat{\mathcal{K}} \neq \mathcal{K}$, e.g. $\mathcal{K}$ may include the belief that only the last six months of data are relevant to the quantile today; yet $\hat{\mathcal{K}}$ may be derived from an industry standard that must use at least one year of observed data in $\hat{\mathcal{K}}$;[16] and secondly, $\hat{F}$ is not, typically, the MED even based on $\hat{\mathcal{K}}$, e.g. the execution of firm-wide VaR models for a large commerical bank may present such a formidable time

[15] In practice, the probability α is often predetermined. Frequently it will be set by senior managers or regulators and small or large values corresponding to extreme quantiles are very commonly used. For instance, regulatory market risk capital is based on VaR models with $\alpha = 1\%$ and a risk horizon of 10 trading days.

[16] As is the case under current banking regulations for the use of VaR to estimate risk capital reserves - see Basel Committee on Banking Supervision (1996).

challenge that $\hat{F}$ is based on simplified data generation processes, as discussed in the introduction.

In the presence of model risk the α quantile of the model is not the α quantile of the MED, i.e. $q_\alpha^{\hat{F}} \neq q_\alpha^F$. The model's α quantile $q_\alpha^{\hat{F}}$ is at a different quantile of F and we use the notation $\hat{\alpha}$ for this quantile, i.e. $q_\alpha^{\hat{F}} = q_{\hat{\alpha}}^F$, or equivalently,

$$\hat{\alpha} = F(\hat{F}^{-1}(\alpha)). \tag{5}$$

In the absence of model risk $\hat{\alpha} = \alpha$ for every α. Otherwise, we can quantify the extent of model risk by the deviation of $\hat{\alpha}$ from α, i.e. the distribution of the quantile probability errors

$$e(\alpha|F, \hat{F}) = \hat{\alpha} - \alpha. \tag{6}$$

If the model suffers from a systematic, measurable bias at the α quantile then the mean error $\bar{e}(\alpha|F, \hat{F})$ should be significantly different from zero. A significant and positive (negative) mean indicates a systematic over (under) estimation of the α quantile of the MED. Even if the model is unbiased it may still lack efficiency, i.e. the dispersion of $e(\alpha|F, \hat{F})$ may be high. Several measures of dispersion may be used to quantify the efficiency of the model. In section 4 we use the the root mean squared error (RMSE) but the standard deviation of the errors, the mean absolute error (MAE) or the range are common alternatives.[17]

We now regard $\hat{\alpha} = F(\hat{F}^{-1}(\alpha))$ as a random variable with a distribution that is generated by our two sources of model risk, i.e. model choice and parameter uncertainty. Because $\hat{\alpha}$ is a probability it has range $[0, 1]$; so we may approximate its distribution by any distribution with support $[0, 1]$. The most general distribution of this type is the generalized beta distribution $\mathcal{GB}(a, b, c)$ which has density (1).

Assuming $\hat{\alpha} \sim \mathcal{GB}(a, b, c)$, the α quantile of our model, adjusted for model risk, becomes a GBG random variable:

$$Q(\alpha|F, \hat{F}) = F^{-1}(\hat{\alpha}), \quad \hat{\alpha} \sim \mathcal{GB}(a, b, c). \tag{7}$$

That is, the model-risk-adjusted quantile $Q(\alpha|F, \hat{F})$ is a random variable whose GBG distribution is generated from the MED F and the parameters of the generalized beta representation of $\hat{\alpha}$. If G_B denotes the distribution of $\mathcal{GB}(a, b, c)$, the distribution function of $Q(\alpha|F, \hat{F})$ is

$$G_F(v; a, b, c) = \Pr(F^{-1}(\hat{\alpha}) \leq v) = G_B(F(v); a, p, q), \quad v \in \mathcal{R}, \tag{8}$$

[17] If the model has a significant bias the MAE, RMSE and range should be applied with caution because they include the bias. Anyway, when the bias is large it is better that the model be inefficient: the worst possible case is an efficient but biased model.

and, denoting by f the density of F, the mean $E[Q(\alpha|F,\hat{F})]$ of $Q(\alpha|F,\hat{F})$ is given by

$$E[Q(\alpha|F,\hat{F})] = B(a,b)^{-1}c\int_{\mathcal{R}} vf(v)F(v)^{ac-1}[1-F(v)^c]^{b-1}dv. \tag{9}$$

This mean quantifies any systematic bias in the quantile estimates: e.g. if the MED has heavier tails than the model then extreme quantiles $q_\alpha^{\hat{F}}$ will be biased: if α is close to zero then $E[Q(\alpha|F,\hat{F})] > q_\alpha^F$ and if α is close to one then $E[Q(\alpha|F,\hat{F})] < q_\alpha^F$. This bias can be removed by adding the difference $q_\alpha^F - E[Q(\alpha|F,\hat{F})]$ to the model's α quantile $q_\alpha^{\hat{F}}$ so that the bias-adjusted quantile has expectation q_α^F.

Unfortunately, analytic expressions for $E[Q(\alpha|F,\hat{F})]$ are only available for some particular choices of F and some values of a, b and c. However, it is always possible to obtain an approximate expression. Upon expanding $F^{-1}(\hat{\alpha})$ in a Taylor series around the point $\hat{\alpha}_m = E(\hat{\alpha})$, we obtain the following a series expansion for $F^{-1}(\hat{\alpha})$ in terms of its derivatives $F^{-1(i)}$, for $i = 1, \ldots, 4$:

$$\begin{aligned} F^{-1}(\hat{\alpha}) &\approx F^{-1}(\hat{\alpha}_m) + F^{-1(1)}(\hat{\alpha}_m)(\hat{\alpha}-\hat{\alpha}_m) + \frac{1}{2}F^{-1(2)}(\hat{\alpha}_m)(\hat{\alpha}-\hat{\alpha}_m)^2 \\ &+ \frac{1}{6}F^{-1(3)}(\hat{\alpha}_m)(\hat{\alpha}-\hat{\alpha}_m)^3 + \frac{1}{24}F^{-1(4)}(\hat{\alpha}_m)(\hat{\alpha}-\hat{\alpha}_m)^4. \end{aligned} \tag{10}$$

Taking expectations in (10) yields

$$\begin{aligned} E[Q(\alpha|F,\hat{F})] &\approx F^{-1}(\hat{\alpha}_m) + \frac{1}{2}F^{-1(2)}(\hat{\alpha}_m)\sigma_{\hat{\alpha}}^2 \\ &+ \frac{1}{6}F^{-1(3)}(\hat{\alpha}_m)\gamma_1(\hat{\alpha})\sigma_{\hat{\alpha}}^3 + \frac{1}{24}F^{-1(4)}(\hat{\alpha}_m)\gamma_2(\hat{\alpha})\sigma_{\hat{\alpha}}^4, \end{aligned} \tag{11}$$

where $\sigma_{\hat{\alpha}}^2 = E[(\hat{\alpha}-\hat{\alpha}_m)^2]$, $\gamma_1(\hat{\alpha}) = \sigma_{\hat{\alpha}}^{-3}E[(\hat{\alpha}-\hat{\alpha}_m)^3]$, and $\gamma_2(\hat{\alpha}) = \sigma_{\hat{\alpha}}^{-4}E[(\hat{\alpha}-\hat{\alpha}_m)^4]$.

The bias-adjusted α quantile estimate could still be far away from the maximum entropy α quantile: the more dispersed the distribution of $Q(\alpha|F,\hat{F})$, the greater the potential for $q_\alpha^{\hat{F}}$ to deviate from q_α^F. Because risk estimates are typically constructed to be conservative, we introduce an *uncertainty buffer* to the bias-adjusted α quantile by adding a quantity equal to the difference between the mean of $Q(\alpha|F,\hat{F})$ and $G_F^{-1}(y)$, the y% quantile of $Q(\alpha|F,\hat{F})$, to the bias-adjusted α quantile estimate. This way, we become $(1-y)$% confident that the model-risk-adjusted α quantile is no less than q_α^F.

Finally, our point estimate for the model-risk-adjusted α quantile becomes:

$$q_\alpha^{\hat{F}} + \overbrace{\{q_\alpha^F - E[Q(\alpha|F,\hat{F})]\}}^{\text{bias adjustment}} + \overbrace{\{E[Q(\alpha|F,\hat{F})] - G_F^{-1}(y)\}}^{\text{uncertainty buffer}} = q_\alpha^{\hat{F}} + q_\alpha^F - G_F^{-1}(y). \tag{12}$$

The total model risk adjustment to the quantile estimate is thus $q_\alpha^F - G_F^{-1}(y)$, and the computation of $E[Q(\alpha|F,\hat{F})]$ could be circumvented if the decomposition into bias and uncertainty components is not required.

The confidence level $1 - y$ reflects a penalty for model risk which is a matter for subjective choice. When X denotes daily P&L and α is small (e.g. 1%), typically all three terms on the right hand side of (12) will be negative. But the α% daily VaR is *minus* the α quantile, so the model-risk-adjusted VaR estimate is $-q_\alpha^{\hat{F}} - q_\alpha^F + G_F^{-1}(y)$. The total adjustment to VaR is $G_F^{-1}(y) - q_\alpha^F$. This will be positive unless, in repeated observations, VaR estimates are typically much greater than the benchmark VaR. In that case there should be a negative bias adjustment, and this could be large enough to outweigh the uncertainty buffer, especially when y is large, i.e. when we require only a low degree of confidence for the model-risk-adjusted VaR to exceed the benchmark VaR.

4 Empirical Illustration

We now describe an experiment in which a portfolio's returns are simulated based on a known data generation process. This allows us to control the degree of VaR model risk and to demonstrate that our framework yields intuitive and sensible results for the bias and inefficiency adjustments described above. We also validate the accuracy of our approximation (11).

It is widely accepted that of all the parsimonious discrete-time variance processes the asymmetric GARCH class is most useful for capturing the volatility clustering that is commonly observed in financial asset returns. Thus, we shall assume that our MED for the returns X_t at time t is $\mathcal{N}(0, \sigma_t^2)$, where σ_t^2 follows an asymmetric GARCH process. First the return x_t from time t to $t+1$ and its variance σ_t^2 are simulated using:

$$\sigma_t^2 = \omega + \alpha(x_{t-1} - \lambda)^2 + \beta\sigma_{t-1}^2, \quad x_t|\mathcal{I}_t \sim \mathcal{N}(0, \sigma_t^2), \tag{13}$$

where $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta \leq 1$ and $\mathcal{I}_t = (x_{t-1}, x_{t-2}, \dots)$.[18] For the simulated returns the parameters of (13) are assumed to be:

$$\omega = 1.5 \times 10^{-6}, \alpha = 0.04, \lambda = 0.005, \beta = 0.95, \tag{14}$$

and so the steady-state annualized volatility of the portfolio return is 25%.[19] Then the MED at time t is $F_t = F(X_t|\mathcal{K}_t)$, i.e. the conditional distribution of the return X_t given the state of knowledge $\mathcal{K}_t$, which comprises the observed returns $\mathcal{I}_t$ and the knowledge that $X_t|\mathcal{I}_t \sim \mathcal{N}(0, \sigma_t^2)$.

At time t, a VaR model provides a forecast $\hat{F}_t = \hat{F}(X_t|\hat{\mathcal{K}}_t)$ where $\hat{\mathcal{K}}$ comprises $\mathcal{I}_t$ plus the model $X_t|\mathcal{I}_t \sim \mathcal{N}(0, \hat{\sigma}_t^2)$. We now consider three different models for $\hat{\sigma}_t^2$. The first model has the correct choice of model but uses incorrect parameter values: instead of (14)

[18] We employ the standard notation α for the GARCH return parameter here; this should not be confused with the notation α for the quantile of the returns distribution, which is also standard notation in the VaR model literature.

[19] The steady-state variance is $\bar{\sigma}^2 = (\omega + \alpha\lambda^2)/(1 - \alpha - \beta)$ and for the annualization we have assumed returns are daily, and that there are 250 business days per year.

the fitted model is:

$$\hat{\sigma}_t^2 = \hat{\omega} + \hat{\alpha}(x_{t-1} - \hat{\lambda})^2 + \hat{\beta}\hat{\sigma}_{t-1}^2, \tag{15}$$

with

$$\hat{\omega} = 2 \times 10^{-6}, \hat{\alpha} = 0.0515, \hat{\lambda} = 0.01, \hat{\beta} = 0.92. \tag{16}$$

The steady-state volatility estimate is therefore correct, but since $\hat{\alpha} > \alpha$ and $\hat{\beta} < \beta$ the fitted volatility process is more 'jumpy' than the simulated variance generation process. In other words, compared with σ_t, $\hat{\sigma}_t$ has a greater reaction but less persistence to innovations in the returns, and especially to negative returns since $\hat{\lambda} > \lambda$.

The other two models are chosen because they are commonly adopted by financial institutions, having been popularized by the 'RiskMetrics' methodology introduced by JP Morgan in the mid-1990's – see RiskMetrics (1997). The second model uses a simplified version of (13) with:

$$\hat{\omega} = \hat{\lambda} = 0, \hat{\alpha} = 0.06, \hat{\beta} = 0.94. \tag{17}$$

This is the RiskMetrics exponentially weighted moving average (EWMA) estimator in which a steady-state volatility is not defined. The third model is the RiskMetrics 'Regulatory' estimator in which:

$$\hat{\alpha} = \hat{\lambda} = \hat{\beta} = 0, \hat{\omega} = \frac{1}{250}\sum_{i=1}^{250} x_{t-i}^2. \tag{18}$$

A time series of 10,000 returns $\{x_t\}_{t=1}^{10,000}$ is simulated from the 'true' model (13) with parameters (14). Then, for each of the three models defined above we use this time series to (a) estimate the daily VaR, which when expressed as a percentage of the portfolio value is given by $\Phi^{-1}(\alpha)\hat{\sigma}_t$, and (b) compute the probability $\hat{\alpha}_t$ associated with this quantile under the simulated returns distribution $F_t = F(X_t|\mathcal{K}_t)$. Because $\Phi^{-1}(\hat{\alpha}_t)\sigma_t = \Phi^{-1}(\alpha)\hat{\sigma}_t$, this is given by

$$\hat{\alpha}_t = \Phi\left[\Phi^{-1}(\alpha)\frac{\hat{\sigma}_t}{\sigma_t}\right]. \tag{19}$$

Taking $\alpha = 1\%$ for illustration, the empirical densities of the three VaR estimates are depicted in Figure 1. The horizontal scale is the percentage VaR (i.e. $\Phi^{-1}(\alpha)\hat{\sigma}_t$) multiplied by 100. The AGARCH VaR is less variable than the EWMA VaR, and the Regulatory VaR has a multi-modal distribution, a feature that results from the lack of risk-sensitivity of this model.[20] Now, for each VaR model, we use the true (i.e. simulated) distribution to estimate $\hat{\alpha}$ at every time point in the simulations, using (19). For $\alpha = 0.1\%, 1\%$ and 5%, Table 1, reports the mean and standard deviation of $\hat{\alpha}$ and the RMSE between $\hat{\alpha}$ and α. The closer $\hat{\alpha}$ is to α, the smaller the RMSE and the less model risk there is in the VaR model. The Regulatory model yields an $\hat{\alpha}$ with the highest RMSE, for every α, so this has the greatest degree of model risk. The AGARCH model, which we already know has

[20]Specifically, a single extreme negative return makes the VaR jump to a level which is sustained for exactly 250 days, and then the VaR jumps down again when this return drops out of the in-sample period.

Figure 1: Density of daily VaR estimates ($\alpha = 1\%$).

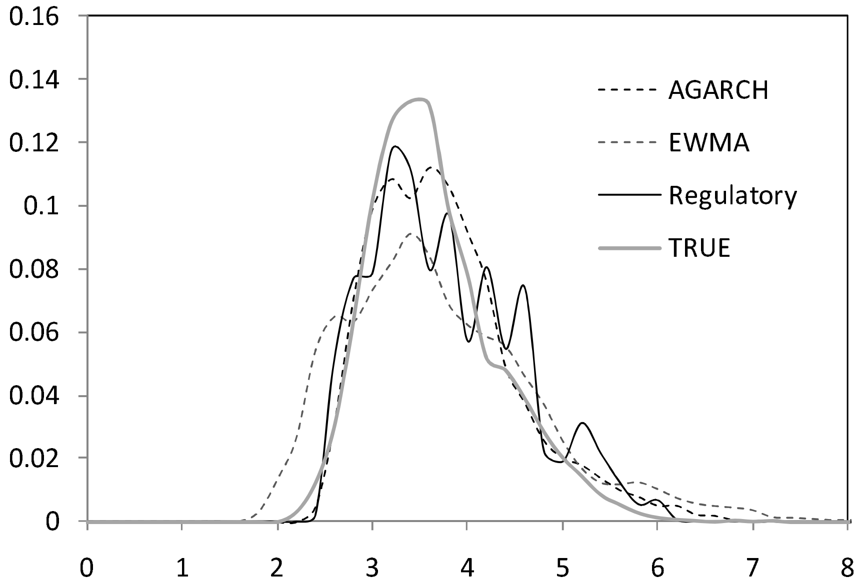
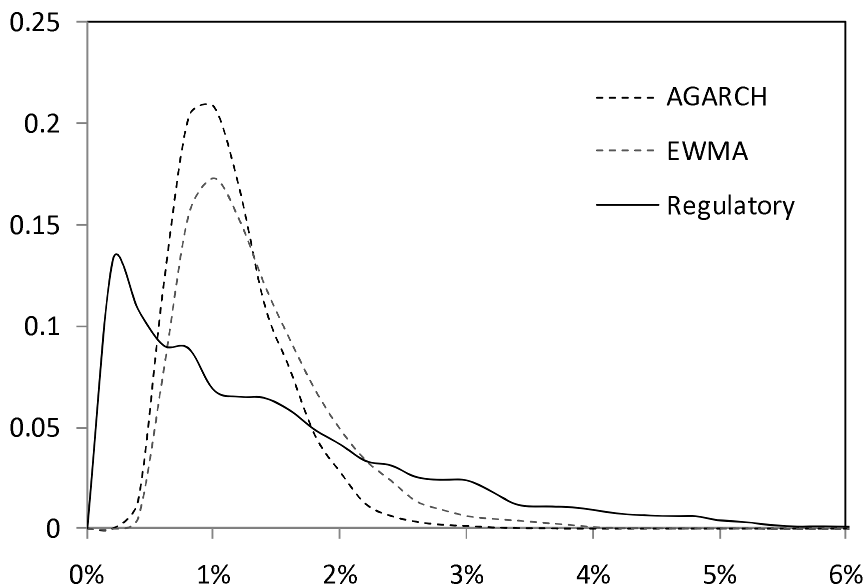


Figure 2: Density of quantile probabilities ($\alpha = 1\%$).



the least model risk of the three, produces a distribution for $\hat{\alpha}$ that has mean closest to the true α and the smallest RMSE. These observations are supported by Figure 2, which depicts the empirical distribution of $\hat{\alpha}$ and Figure 3, which shows the empirical densities of the model-risk-adjusted VaR estimates $F^{-1}(\hat{\alpha})$ (here the horizontal axis is the same as Figure 1, i.e. the VaR as a percentage of the portfolio value, multiplied by 100). Both these figures take $\alpha = 1\%$ for illustration.

Figure 3: Density of model-risk-adjusted daily VaR ($\alpha = 1\%$).

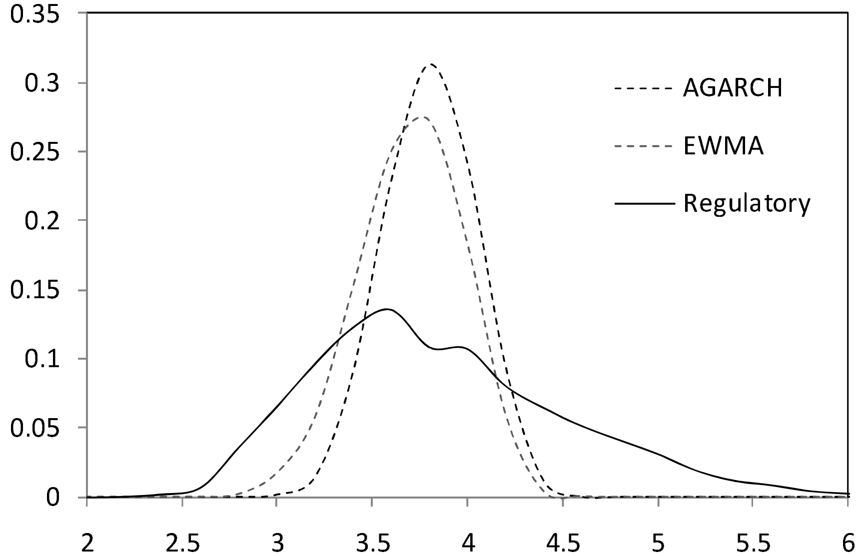


Table 1: Sample statistics for quantile probabilities

		AGARCH	**EWMA**	**Regulatory**
0.10%	Mean	0.11%	0.16%	0.23%
	Stdev	0.0007	0.0013	0.0035
	RMSE	0.07%	0.14%	0.37%
1%	Mean	1.03%	1.25%	1.34%
	Stdev	0.0042	0.0059	0.0119
	RMSE	0.42%	0.64%	1.22%
5%	Mean	4.97%	5.44%	5.27%
	Stdev	0.0103	0.0123	0.0268
	RMSE	1.03%	1.31%	2.66%

Table 2: Values of $(\hat{a}, \hat{b})$ for the distributions given in Table 1.

α		**AGARCH**	**EWMA**	**Regulatory**
0.10%	$\hat{a}$	2.35	1.50	0.43
	$\hat{b}$	2145.39	959.21	188.35
1%	$\hat{a}$	5.92	4.43	1.24
	$\hat{b}$	570.98	350.90	91.04
5%	$\hat{a}$	22.10	18.29	3.61
	$\hat{b}$	422.04	318.15	64.78

We fit a beta distribution to $\hat{\alpha}$, for each model, and show that the approximation (11) is valid. Table 2 reports the method of moment beta parameter estimates, denoted $(\hat{a}, \hat{b})$, that are computed using the means and standard deviations given in Table 1. Since $\hat{a} << \hat{b}$ all beta distributions have a large positive skew. Both $\hat{a}$ and $\hat{b}$ are inversely proportional to α, and to the degree of model risk. Thus, at one extreme we have the highly-peaked beta density of the AGARCH model when $\alpha = 5\%$, and at the other we have the 'L'-shaped density of the Regulatory model when $\alpha = 0.1\%$.

Next we compare the empirical mean with the mean of our beta normal distribution for the adjusted VaR. This is computed using numerical integration of (9) or via the analytic approximation (11). All three means are subject to error, albeit in slightly different ways, but the errors are small relative to the VaR model risk. The numerical and analytic computations are based on the beta parameter estimates in Table 2. Note that $d\phi(z)/dz = -z\phi(z)$ and that the derivatives of $\Phi^{-1}(z)$ in (11) are given by:

$$\begin{aligned}
\Phi^{-1(1)}(z) &= \phi(\Phi^{-1}(z))^{-1}, \\
\Phi^{-1(2)}(z) &= \Phi^{-1}(z)[\phi(\Phi^{-1}(z))]^{-2}, \\
\Phi^{-1(3)}(z) &= [1 + 2\left\{\Phi^{-1}(z)\right\}^2][\phi(\Phi^{-1}(z))]^{-3}, \\
\Phi^{-1(4)}(z) &= \Phi^{-1}(z)[7 + 6\left\{\Phi^{-1}(z)\right\}^2][\phi(\Phi^{-1}(z))]^{-4}.
\end{aligned}$$

Table 3: Mean of the model-risk-adjusted VaR distribution

α	Mean	**AGARCH**	**EWMA**	**Regulatory**
0.10%	(Emp.)	4.919	4.793	4.961
	(9)	4.953	4.832	-
	(11)	4.988	5.017	1.292
1%	(Emp.)	3.703	3.608	3.735
	(9)	3.722	3.603	3.729
	(11)	3.710	3.614	4.067
5%	(Emp.)	2.618	2.551	2.641
	(9)	2.626	2.227	2.643
	(11)	2.618	2.552	2.654

The results are reported in Table 3. The approximation (11) is best for the AGARCH model, which has the smallest degree of model risk. The least accurate approximation is for the Regulatory VaR model at extreme quantiles, when the bias in the VaR estimates becomes quite large. At the 0.1% quantile, where the Regulatory model has a value of $\hat{a}$ less than 1, it becomes impossible to integrate (9) using standard numerical methods.

A point estimate for model-risk-adjusted VaR (RaVaR, for short) is computed using

(12). Both RaVaR and the benchmark VaR (BVaR, for short) depend on the time they are measured. For illustration, we select a point when the simulated volatility is at its steady-state value of 25% – so the BVaR is 4.886%, 3.678% and 2.601% at the 0.1%, 1% and 5% levels, respectively. Drawing at random from the points when the simulated volatility was 25%, we obtain AGARCH, EWMA and Regulatory volatility forecasts of 27.00%, 23.94% and 28.19% respectively.[21] These volatilities determine the VaR estimates that we shall now adjust for model risk.

Table 4: Components of the model risk adjustment

α		**AGARCH**	**EWMA**	**Regulatory**
0.1%	Mean	4.919	4.793	4.961
	Bias	-0.033	0.093	-0.075
	Quantile	4.447	4.177	3.912
	Uncert. Buffer	0.472	0.615	1.049
1%	Mean	3.703	3.608	3.735
	Bias	-0.025	0.070	-0.056
	Quantile	3.348	3.145	2.945
	Uncert. Buffer	0.355	0.463	0.789
5%	Mean	2.618	2.551	2.641
	Bias	-0.017	0.050	-0.040
	Quantile	2.366	2.224	2.082
	Uncert. Buffer	0.252	0.327	0.559

Table 4 summarizes the bias and the uncertainty buffer, for different levels of α, based on the empirical distribution of $Q(\alpha|F, \hat{F})$.[22] It reveals a general tendency for the EWMA model to slightly underestimate VaR and the other models to slightly overestimate VaR. Yet the bias is relatively small, since all models assume the same normal form as the MED and the only difference between them is their volatility forecast. Although the bias tends to increase as α decreases it is not significant for any model.[23] Beneath the bias we report the 5% quantile of the model-risk-adjusted VaR distributions, since we shall first compute the RaVaR so that it is no less than the BVaR with 95% confidence.

Following the framework introduced in the previous section we now define:

$$\text{RaVaR}(y) = \text{VaR} + \overbrace{(\text{BVaR} - E[Q(\alpha|F, \hat{F})])}^{\text{bias adjustment}} + \overbrace{(E[Q(\alpha|F, \hat{F})] - G_F^{-1}(y))}^{\text{uncertainty buffer}}.$$

[21]So the AGARCH and Regulatory models overestimated BVaR at this point and the EWMA model underestimated BVaR; of course, this was not the case every time the simulated volatility was 25%.

[22]Similar results based on the fitted distributions are not reported for brevity.

[23]Standard errors of $Q(\alpha|F, \hat{F})$ are not reported, for brevity. They range between 0.157 for the AGARCH at 5% to 0.891 for the Regulatory model at 0.1%, and are directly proportional to the degree of model risk just like the standard errors on the quantile probabilities given in Table 1.

Table 5 sets out the RaVaR computation for $y = 5\%$. The model's volatility forecasts are in the first row and the corresponding VaR estimates are in the first row of each cell, for $\alpha = 0.1\%$, 1% and 5% respectively. The (small) bias is corrected by adding the bias from Table 4 to each VaR estimate. The main source of model risk here concerns the potential for a large (positive or negative) errors in the quantile probabilties, i.e. the dispersion of the densities in Figure 2. To adjust for this we add to the bias-adjusted VaR an uncertainty buffer equal to the difference between the BVaR and the 5% quantile given in Table 4.[24] This gives the RaVaR estimates shown in the third row of each cell.

Table 5: Computation of 95% RaVaR

α		**AGARCH**	**EWMA**	**Regulatory**
	Volatility	27.00%	23.94%	28.19%
0.10%	VaR	5.277	4.678	5.509
	Bias Adj. VaR	5.244	4.772	5.434
	RaVaR	5.716	5.387	6.483
1%	VaR	3.972	3.522	4.147
	Bias Adj. VaR	3.948	3.592	4.091
	RaVaR	4.303	4.055	4.880
5%	VaR	2.809	2.490	2.932
	Bias Adj. VaR	2.791	2.540	2.892
	RaVaR	3.043	2.867	3.451

Since risk capital is a multiple of VaR, the percentage increase resulting from replacing VaR by RaVaR(y) is:

$$\%\text{ risk capital increase} = \frac{\text{BVaR} - G_F^{-1}(y)}{\text{VaR}}\ . \tag{20}$$

The penalty (20) for model risk depends on α, except in the case that both the MED and VaR model are normal, and on the confidence level $(1-y)\%$.

Table 6 reports the percentage increase in risk capital due to model risk when RaVaR is no less than the BVaR with $(1-y)\%$ confidence. As expected, this turns out to be directly proportional to the degree of model risk in the VaR model. At 95% confidence, the first row of the table shows that risk capital based on the AGARCH model would be increased by about 8%, increased by about 15% in the EWMA model and about 17.5% in the Regulatory model. The other rows in Table 6 give the amount of additional risk captial that would be required at other reasonable confidence levels. The add-on for VaR model risk increases with the degree of model risk, and with the degree of confidence that the regulator requires for the model-risk-adjusted VaR to be at least as great as the benchmark

[24]Recall, the BVaR is 4.886%, 3.678% and 2.601% at the 0.1%, 1% and 5% levels, respectively.

Table 6: Percentage increase in risk capital from model risk adjustment of VaR

$1-y$	**AGARCH**	**EWMA**	**Regulatory**
95%	8.33%	15.15%	17.68%
90%	6.05%	12.62%	14.16%
85%	4.47%	10.99%	11.37%
80%	3.19%	9.57%	8.81%
75%	2.25%	8.38%	6.68%

VaR. Note that the risk capital estimate could require the greatest upwards adjustment when based on EWMA VaR. Indeed, if we require only a low degree of confidence (80% or less) that the RaVaR exceeds the BVaR, the additional capital required for a EWMA VaR model exceeds that required for a Regulatory VaR model. This is because the EWMA VaR is exceptionally low at the time of the adjustment.

5 Summary and Conclusions

National regulators will very soon require that banks set aside additional reserves to cover all sources of model risk in the internal models used to compute the market risk capital charge. This paper concerns VaR model risk, i.e. the risk of producing inaccurate VaR estimates because of an inappropriate choice of VaR model (e.g. historical simulation or parametric analytic) and/or inaccuracy in the VaR model parameter estimates. We present a rigorous statistical methodology that provides an elegant and practical solution to the problem of quantifying the regulatory capital that should be set aside to cover this type of model risk.

We argue that there is no better choice of model risk benchmark than a maximum entropy distribution since, by definition, this embodies the entirety of information and beliefs, no more and no less. In the context of the model risk capital charge, the benchmark should be specified by the regulator. Then VaR model risk is assessed using a top-down approach, by comparing two time series: one for the benchmark VaR, which is derived directly from the bank's total daily P&L, and the other for the bank's aggregate daily VaR figures, upon which their market risk capital charge is based. The latter is typically derived using a computationally intensive bottom-up approach, with many approximations and simplifications being applied.

The main ideas are as follows: in the presence of model risk an α quantile is at a different quantile of the benchmark model, and has an associated tail probability under the benchmark that is stochastic. Modelling this uncertainty as generally as possible, the model-risk-adjusted quantile becomes a generalized beta generated random variable, and its distribution quantifies the bias and uncertainty due to model risk. A significant model risk bias arises if the VaR estimates produced by the bank for the purposes of risk capital charge calculation tend to be consistently above or below the VaR estimates obtained by applying the benchmark model to their aggregate daily P&L. Even when the bank's VaR estimates show no bias, an adjustment for uncertainty is required because the difference between the bank's VaR and the benchmark VaR could vary considerably over time. The bias and uncertainty in the VaR model, relative to the benchmark, determine a risk capital adjustment for model risk whose size will also depend on the confidence level regulators require for the adjusted risk capital to be no less than the risk capital based on the benchmark model. An empirical example that considers three VaR models with controlled degrees of model risk has been used to illustrate the framework.

Further research is required in conjunction with both regulators and banks – on the development of benchmark VaR models to apply to aggregate P&L, and on backtesting the model-risk-adjusted estimates for commonly-used VaR models. The extension of our methodology to other quantile-based metrics used in finance, and to conditional VaR in particular, should be straightforward.[25] The fundamental idea of using a stochastic probability to generate another distribution would still hold, and the bias and uncertainty of the generated ditribution could still be used to quantify the effect of model risk. Our work also has potential applications to many other disciplines where quantile risk assessments are used, such as insurance, hydrology, climate change, statistical process control and reliability analysis.

[25] Conditional VaR is also called 'expected tail loss' or 'expected shortfall' by a variety of authors.

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